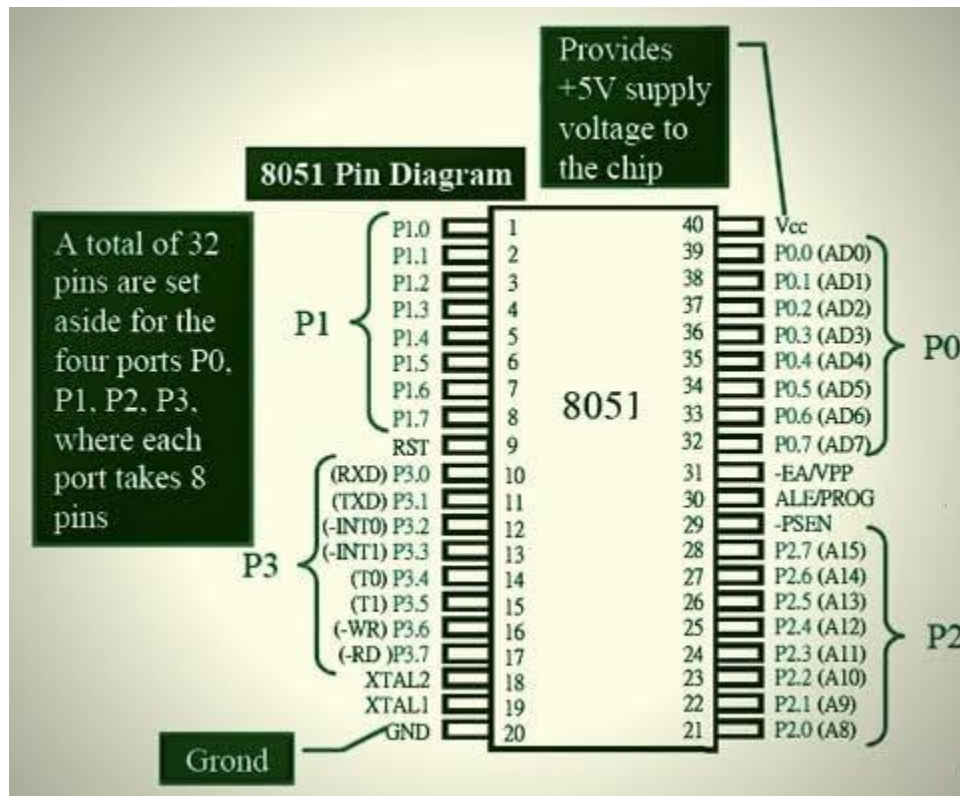


Embedded Computing Principles

Name: Sachin Kumar

Admission No. 25SECE1030112

8051 Microcontroller – Summary



The 8051 microcontroller is a popular 8-bit microcontroller originally developed by Intel. It is widely used in embedded systems, automation, control applications, and educational projects because of its simple architecture and programming features. The 8051 combines a processor, memory, input/output ports, timers, interrupts, and serial communication facilities on a single chip.

The classic 8051 has an 8-bit CPU, 4 KB of ROM, 128 bytes of internal RAM, and 32 programmable I/O pins. These 32 pins are divided into four 8-bit ports: Port 0, Port 1, Port 2, and Port 3. Port 0 and Port 2 can also be used for external memory interfacing, while Port 3 provides several alternate functions such as serial communication and external interrupts.

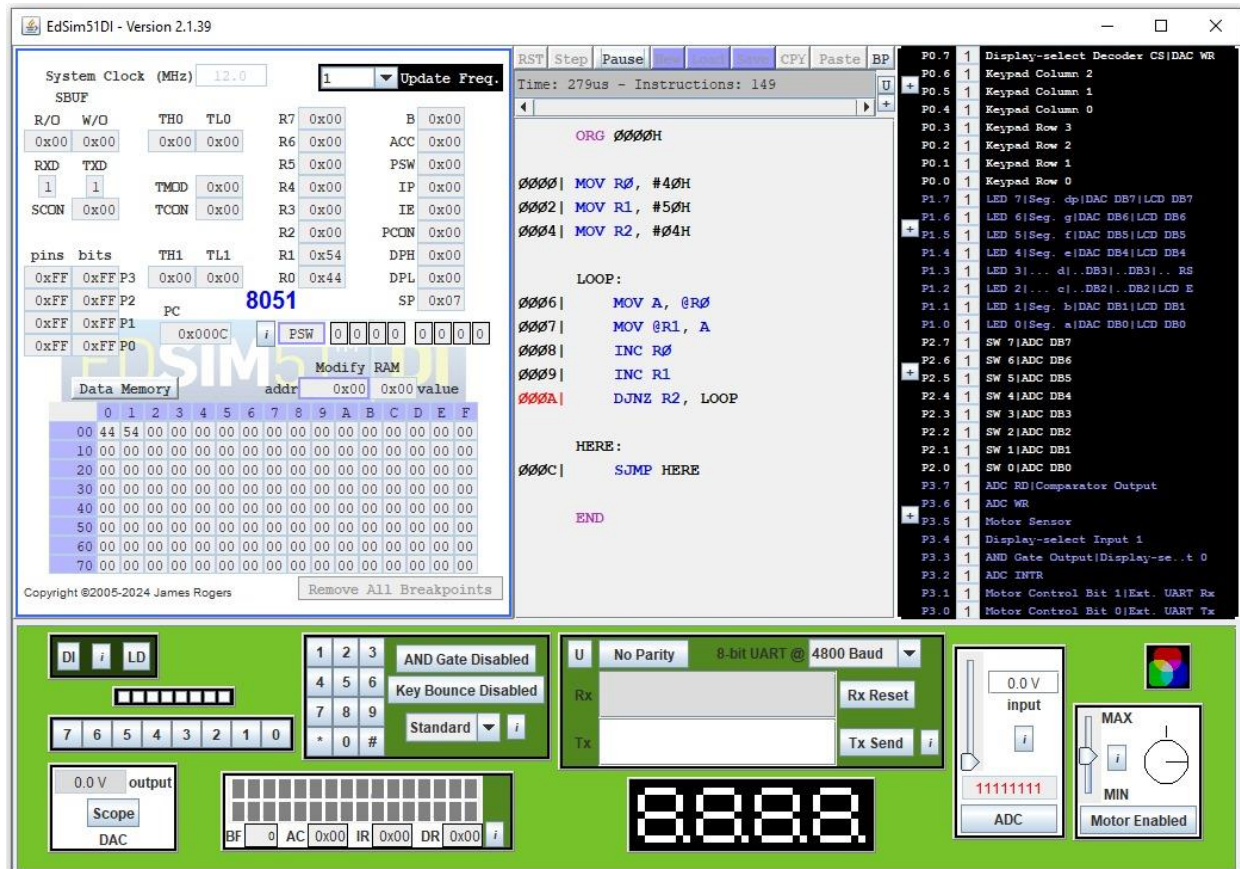
The 8051 contains two 16-bit timers/counters, Timer 0 and Timer 1. These can be used for generating time delays, counting external events, and controlling various timing operations. The microcontroller also supports interrupts, which allow it to respond quickly to important external or internal events.

For communication, the 8051 provides a full-duplex serial communication interface using the RXD and TXD pins. It also contains several Special Function Registers (SFRs) that are used to control and configure its different operations.

The 8051 supports various addressing modes and has an instruction set for arithmetic, logical, data transfer, branching, and bit manipulation operations. Programs can be written in Assembly language or C.

The 8051 is commonly used for LED control, LCD displays, keyboards, sensors, motors, traffic-light systems, temperature monitoring, and industrial automation. Overall, the 8051 is an important microcontroller for understanding the basic concepts of embedded systems and microcontroller programming.

Q. Write a program to copy 4 data elements from 1 memory block (40H to 45H) to another memory block.



CODE:

```
ORG 0000H
```

MOV R0, #40H

MOV R1, #50H

MOV R2, #04H

LOOP:

MOV A, @R0

MOV @R1, A

INC R0

INC R1

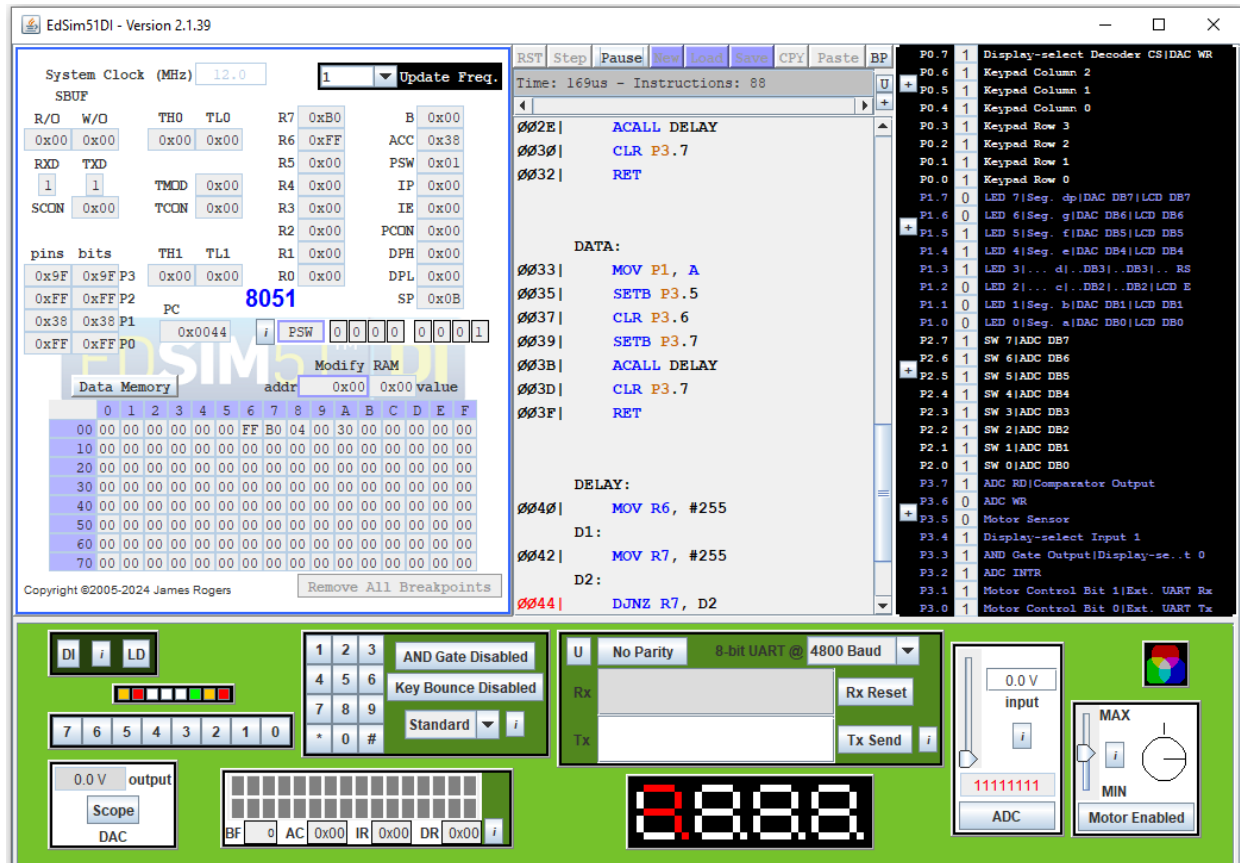
DJNZ R2, LOOP

HERE:

SJMP HERE

END

Q. Write a program to blink the lcd display (2 digit of lcd display) in edsim51



CODE:

ORG 0000H

MOVA, #38H

ACALL COMMAND

MOVA, #0CH

ACALL COMMAND

MOVA, #01H

ACALL COMMAND

MOVA, #80H

ACALL COMMAND

MOVA, #'2'

ACALL DATA

MOVA, #'5'

ACALL DATA

BLINK:

ACALL DELAY

MOVA, #08H

ACALL COMMAND

ACALL DELAY

MOVA, #0CH

ACALL COMMAND

SJMP BLINK

COMMAND:

MOV P1, A

CLR P3.5

CLR P3.6

SETB P3.7

ACALL DELAY

CLR P3.7

RET

DATA:

MOV P1, A

SETB P3.5

CLR P3.6

SETB P3.7

ACALL DELAY

CLR P3.7

RET

DELAY:

MOV R6, #255

D1:

MOV R7, #255

D2:

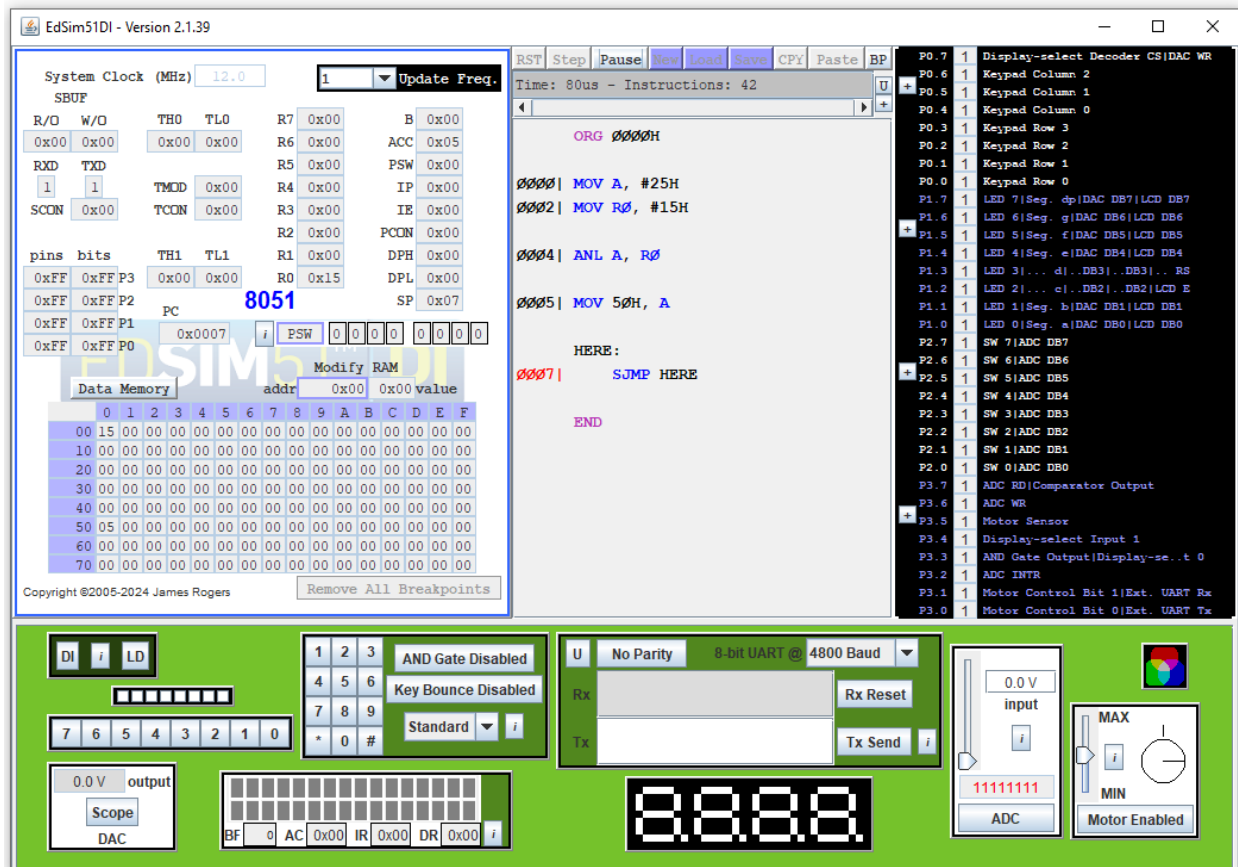
DJNZ R7, D2

DJNZ R6, D1

RET

END

Q. Write a program to perform AND operation



CODE:

ORG 0000H

MOVA, #25H

MOV R0, #15H

ANL A, R0

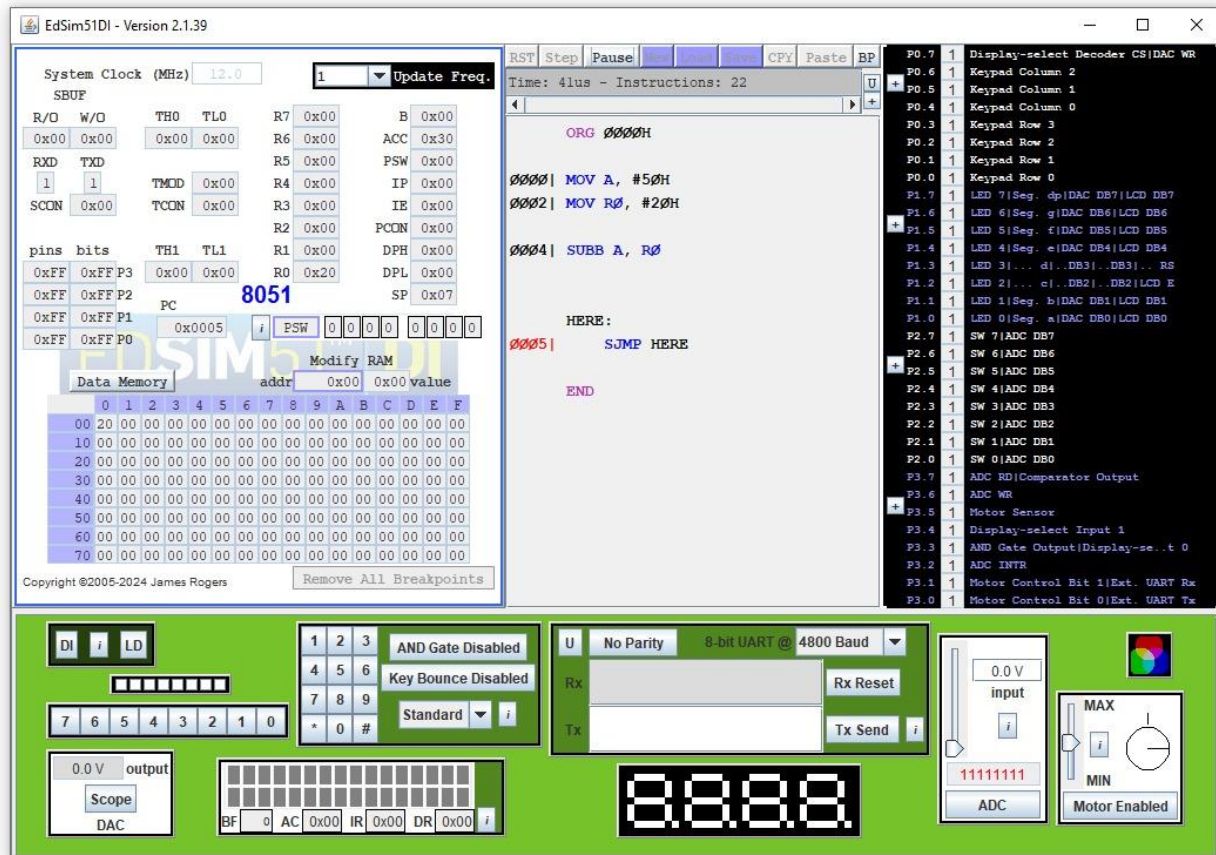
MOV 50H, A

HERE:

SJMP HERE

END

Q. Write a program to perform subtraction 20h FROM 50H



CODE:

ORG 0000H

MOV A, #50H

MOV R0, #20H

SUBB A, R0

HERE:

SJMP HERE

END